

INHERITANCE GENETICS

Module map

- Module 09: Inheritance Genetics
- Connected lab: lab-09_inheritance-genetics.md
- Use this deck as the visual path through the module's evidence and retrieval work.

Module 09

Inheritance Genetics

1

Alleles and
genotype

2

Mendelian
patterns

3

Punnett-
square
prediction

4

Beyond
simple
dominance

5

Pedigrees
and
probability

Lab evidence

lab-09_inheritance-genetics.md

INHERITANCE GENETICS

Learning objectives

- Distinguish gene, allele, genotype, and phenotype.
- Use segregation to explain monohybrid inheritance.
- Build and interpret Punnett-square probabilities.
- Recognize inheritance patterns beyond complete dominance.
- Use pedigree evidence to reason about traits.

OBJECTIVE 1

Distinguish gene, allele, genotype, and phenotype.

OBJECTIVE 2

Use segregation to explain monohybrid inheritance.

OBJECTIVE 3

Build and interpret Punnett-square probabilities.

OBJECTIVE 4

Recognize inheritance patterns beyond complete dominance.

OBJECTIVE 5

Use pedigree evidence to reason about traits.

INHERITANCE GENETICS

Topic sequence

- Alleles and genotype: Alleles are gene variants that contribute to genotype and phenotype.
- Mendelian patterns: Mendelian inheritance predicts patterns when alleles segregate into gametes.
- Punnett-square prediction: Punnett squares model probability, not guaranteed outcomes.
- Beyond simple dominance: Incomplete dominance, codominance, polygenic traits, and environment complicate simple patterns.
- Pedigrees and probability: Pedigrees use family evidence to infer inheritance patterns.

01

Alleles and genotype

Alleles are gene variants that contribute to genotype and phenotype.

02

Mendelian patterns

Mendelian inheritance predicts patterns when alleles segregate into gametes.

03

Punnett-square prediction

Punnett squares model probability, not guaranteed outcomes.

04

Beyond simple dominance

Incomplete dominance, codominance, polygenic traits, and environment complicate simple patterns.

05

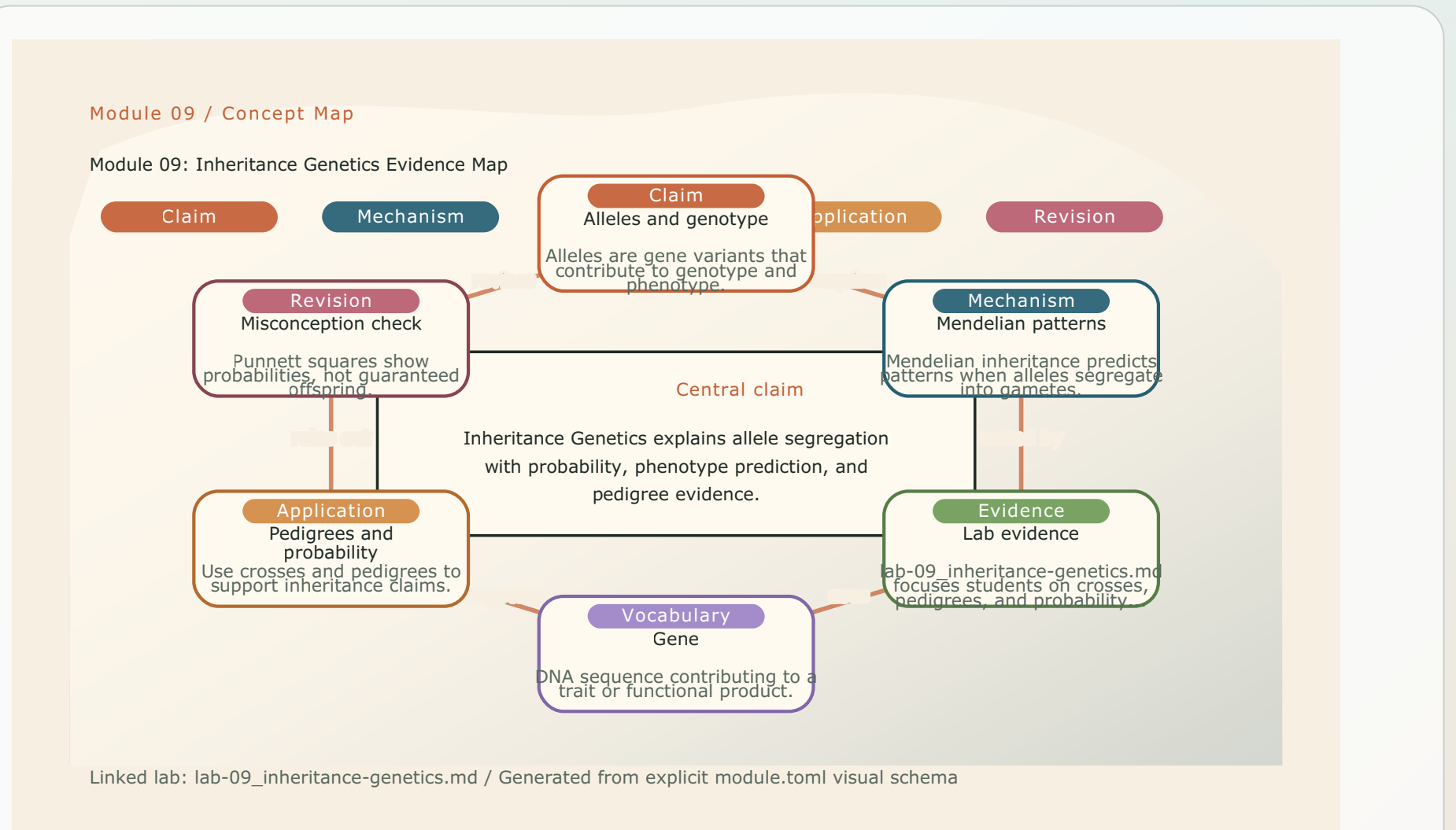
Pedigrees and probability

Pedigrees use family evidence to infer inheritance patterns.

INHERITANCE GENETICS

Module 09: Inheritance Genetics Evidence Map

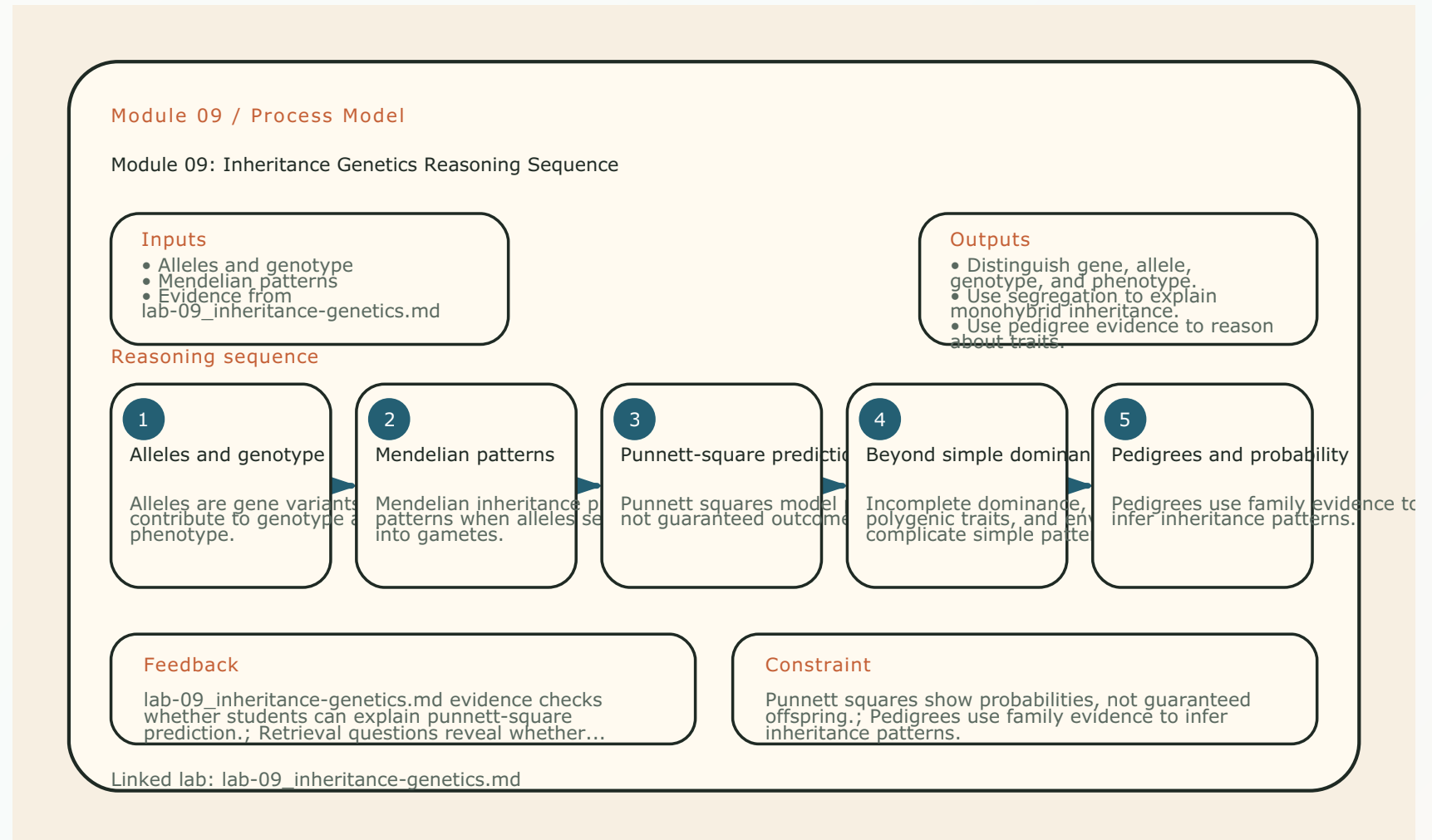
- Module focus: Inheritance Genetics
- Central claim: Inheritance Genetics explains allele segregation with probability, phenotype prediction, and pedigree evidence.
- Key nodes to connect: Alleles and genotype, Mendelian patterns, Lab evidence
- Reasoning move: use 'requires' to explain relationships, not memorize terms.
- Lab connection: lab-09_inheritance-genetics.md supplies an evidence surface for the map.



INHERITANCE GENETICS

Module 09: Inheritance Genetics Reasoning Sequence

- Module focus: Inheritance Genetics
- Trace the model: Alleles and genotype -> Mendelian patterns -> Punnett-square prediction -> Beyond simple dominance
- Inputs become outputs: Alleles and genotype, Mendelian patterns -> Distinguish gene, allele, genotype, and phenotype., Use segregation to explain monohybrid inheritance.
- Feedback check: lab-09_inheritance-genetics.md evidence checks whether students can explain punnett-square prediction.
- Lab connection: lab-09_inheritance-genetics.md gives students a process to observe or test.



INHERITANCE GENETICS

Terms as evidence handles

- Gene: DNA sequence contributing to a trait or functional product.
- Allele: Variant form of a gene.
- Genotype: Allele combination of an organism.
- Phenotype: Observable trait or characteristic.
- Dominant: Allele effect visible with one copy in simple dominance.
- Recessive: Allele effect masked by dominant allele in heterozygotes.

GENE

DNA sequence contributing to a trait or functional product.

ALLELE

Variant form of a gene.

GENOTYPE

Allele combination of an organism.

PHENOTYPE

Observable trait or characteristic.

DOMINANT

Allele effect visible with one copy in simple dominance.

RECESSIVE

Allele effect masked by dominant allele in heterozygotes.

INHERITANCE GENETICS

Lab connection

- Lab file: lab-09_inheritance-genetics.md
- Lab evidence should test or illustrate:
Punnett-square prediction
- Students should leave with a concrete
observation, comparison, or model tied to
the module claim.

QUESTION

Alleles and genotype

EVIDENCE

lab-09_inheritance-
genetics.md

CLAIM

Distinguish gene,
allele, genotype, and
phenotype.

INHERITANCE GENETICS

Contrast check

- Do not stop at: Alleles are gene variants that contribute to genotype and phenotype.
- Stronger explanation: Pedigrees use family evidence to infer inheritance patterns.
- The goal is to move from a named idea to a testable biological explanation.

SURFACE

Alleles are gene variants that contribute to genotype and phenotype.

DEEPER

Pedigrees use family evidence to infer inheritance patterns.

INHERITANCE GENETICS

Module 09: Inheritance Genetics Retrieval and Lab Check

- Module focus: Inheritance Genetics
- Retrieval prompt: How are genes and alleles related?
- Answer check: Distinguish gene, allele, genotype, and phenotype.
- Required terms: Gene, Allele, Genotype
- Lab connection: lab-09_inheritance-genetics.md; lab-09_inheritance-genetics.md supplies evidence for crosses, pedigrees, and probability patterns in inheritance.

Module 09 / Retrieval Card

Module 09: Inheritance Genetics

Retrieval and Lab Check

Routine

Cover notes -> answer aloud
-> cite evidence -> revise

1 Cover notes

2 Answer aloud

3 Cite evidence

4 Revise

Q1

How are genes and alleles related?

Check: Distinguish gene, allele, genotype, and phenotype.

Q2

How do genotype and phenotype differ?

Check: Use segregation to explain monohybrid inheritance.

Q3

What does heterozygous mean?

Check: Build and interpret Punnett-square probabilities.

Q4

How does allele segregation explain a monohybrid cross?

Check: Recognize inheritance patterns beyond complete dominance.

Terms to use

Gene, Allele, Genotype, Phenotype, Dominant

Lab connection

lab-09_inheritance-genetics.md supplies evidence for crosses, pedigrees, and probability patterns in...

Intro biology routine: claim, evidence, reasoning, revision.

Teaching note: Pause here. Students answer first without notes, then compare to the checks.

INHERITANCE GENETICS

Practice quiz bridge

- 1. An organism's genotype is its:
- 2. Two heterozygotes are crossed ($Aa \times Aa$). What fraction of offspring are expected to show the recessive trait?
- 3. A heterozygote (Aa) shows the dominant trait because:
- 4. What is a pedigree used for?

QUESTION 1**Key: C**

An organism's genotype is its:

QUESTION 2**Key: A**

Two heterozygotes are crossed ($Aa \times Aa$). What fraction of offspring are expected to show the recessive trait?

QUESTION 3**Key: D**

A heterozygote (Aa) shows the dominant trait because:

QUESTION 4**Key: B**

What is a pedigree used for?

INHERITANCE GENETICS

Synthesis and exit ticket

- Exit claim: explain alleles and genotype using evidence from lab-09_inheritance-genetics.md.
- Must include: Gene, Allele, Genotype.
- Revision prompt: what evidence would change your explanation?

CLAIM

Alleles and genotype

EVIDENCE

lab-09_inheritance-genetics.md

REVISION

What would change your mind?